

Supervisor Progress Report

Label-Private Split Federated Learning Framework

Current implementation status, architecture, training flow, and experimental metrics

1. Current Status

SplitFed phase status: Completed and tested with a real client-server setup. Focus: framework design, implementation progress, validation, performance, and communication cost.

Area	Current implementation
Learning setup	Label-private Split Federated Learning for MIMIC-IV mortality prediction
Client count	3 non-IID client partitions
Client runtime	Local machine
Server runtime	Google Colab
Communication	Cloudflare public tunnel
Aggregation	FedAvg across client encoders after each round
Validation status	All major API calls returned successful 200 OK responses during medium-scale runs

2. Dataset and Client Partitioning

MIMIC-IV was partitioned into three non-IID clinical client groups. This matches the broader HCQ-SFL-IR research context where client-side data remains inside institutional or unit-level boundaries.

Client	Partition	Description
Client 0	Medical	Medical ICU-style patient partition
Client 1	Surgical	Surgical ICU-style patient partition
Client 2	Cardiac	Cardiac ICU-style patient partition

3. Implemented SplitFed Framework

Component	Location	Role
ClientEncoder	Local client side	Converts private ICU features into intermediate activations
ServerTopModel	Remote server side	Receives activations and produces mortality prediction logits

Client keeps	Server keeps
Raw patient records	Server-side prediction model
Mortality labels	Temporary activation-gradient state
Scaler and local preprocessing	FedAvg aggregation state
Client encoder	Server logs and checkpoint state
Local dataloader	Validation prediction endpoint

4. Model Architecture

Model part	Input	Architecture	Output	Parameters
ClientEncoder	266 ICU clinical features	Linear(266->64) + BatchNorm + ReLU + Dropout(0.5) + Linear(64->32) + BatchNorm + ReLU	32-dimensional activation	19,360
ServerTopModel	32-dimensional activation	Linear(32->64) + BatchNorm + ReLU + Dropout(0.3) + Linear(64->1)	Mortality logit	2,305

5. Label-Private Activation-Gradient Flow

1. Client processes a local patient batch.
2. ClientEncoder generates intermediate activations.
3. Client sends activations to the remote server.
4. ServerTopModel returns prediction logits.
5. Client computes loss locally using private mortality labels.
6. Client sends only the gradient signal back to the server.
7. Server backpropagates through the server-side model.
8. Server returns activation gradients to the client.
9. Client updates its encoder locally.
10. After each round, client encoders are aggregated using FedAvg.

Privacy boundary	Status
Raw patient data	Never leaves the client
Labels	Never leave the client
Server receives	Activations and gradient signals only

6. Distributed System Validation

The framework was validated through a real distributed setup, not only a notebook-level simulation. The Colab server successfully handled the following request types:

- Client registration
- Activation forward pass
- Gradient backward pass
- Validation prediction
- FedAvg submission
- Global encoder retrieval

7. Medium-Scale Experiment Setup

Item	Value
Number of clients	3
Training rounds	5
Training batches	200 batches per client per round
Validation	Limited validation batches
Server	Remote Google Colab server
Client data and labels	Kept locally on each client
Aggregation schedule	FedAvg after every round

8. AUROC Progress Across Rounds

Round	Medical AUROC	Surgical AUROC	Cardiac AUROC	Mean AUROC
1	0.9515	0.9636	0.9749	0.9633
2	0.9448	0.9670	0.9749	0.9622
3	0.9449	0.9666	0.9775	0.9630
4	0.9467	0.9662	0.9774	0.9634
5	0.9476	0.9677	0.9781	0.9644

9. Round 5 Detailed Metrics

Client	Loss	AUROC	AUPRC	F1
Medical	0.5507	0.9476	0.8297	0.6838
Surgical	0.3903	0.9677	0.8543	0.7432
Cardiac	0.3147	0.9781	0.8363	0.6859

10. Result Interpretation

- Training remained stable across all three non-IID clients.
- FedAvg completed successfully after every round.
- Mean AUROC stayed consistently high, reaching 0.9644 by round 5.
- The label-private protocol preserved the intended privacy boundary throughout the run.
- Client-specific AUROC values show strong performance across medical, surgical, and cardiac partitions.

11. Activation-Gradient Communication Cost

The estimate below covers the core activation and activation-gradient exchange for the medium-scale 5-round run. It excludes small metadata, validation prediction calls, and model-weight transfer during FedAvg.

Per client per round	Bytes
Activations sent to server	1,638,400
Activation gradients returned to client	1,638,400
Total per client per round	3,276,800

All clients	Bytes
Total communication per round	9,830,400
Total over 5 rounds	49,152,000

12. Supervisor-Level Summary

The implemented phase establishes a working label-private SplitFed foundation for the broader HCQ-SFL-IR / QR-SecaaS research direction. The current system supports real client-server training between local clients and a remote Colab server, keeps both raw MIMIC-IV records and mortality labels client-side, performs activation-gradient communication, aggregates client encoders through FedAvg, and produces stable validation results across three non-IID clinical partitions.